

MBSD Assignment #2 A.Y. 2024/25

Purposes

- Implement the “one pedal controller” as a Simulink model.

To implement the system, follow the description already provided in the Laboratory 1 document considering also eventual assumptions already stated in this latter document. The safety mechanisms are not required in this laboratory.

The Simulink project has to be split into 3 files:

- **Harness.slx**, containing reference models for the controller and plant and test stimuli generation
- **Controller.slx**, containing the controller (to be developed)
- **Plant.slx**, containing the car longitudinal physical model.

The longitudinal physical model and the test stimuli generators shall demonstrate the effectiveness of the implemented controller with respect to the expected functionalities. Consider the plant model description as an example of the comments to be inserted in the Controller Software Unit description report.

Templates of these files are available alongside this document.

The deliverable, composed of the 3 Simulink models and a PDF file obtained by filling the following pages of this document (please delete this first page), has to be provided as a .ZIP file up to **May 4th at 23:59 through “Consegna Elaborati” of the Portale della Didattica**. It shall also contain a brief report explaining the design of the controller, using the following template. It is sufficient that only one of the group members uploads it.

Model-Based Software Design, A.Y. 2024/25

Laboratory 2 Report

Components of the working group (max 2 people)

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- Mohammad Nour Edeen 344035

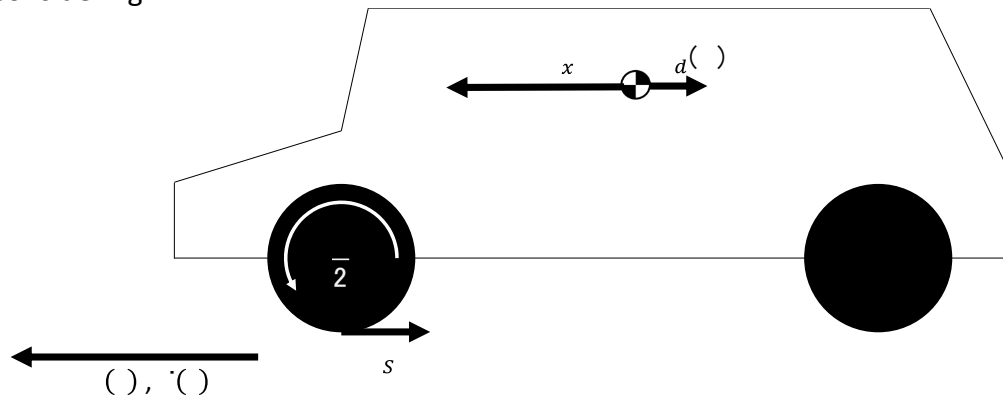
External interfaces of the plant

Name	Direction	Type
Requested_Torque_Nm	Input	CAN
Vehicle_Speed_kmh	Output	CAN
Automatic_Transmission_Selector	Input (from the driver to the controller)	CAN {P, R, N, D, P}
Selected mode/errors	Output (to the driver)	CAN

Equations of the plant

The plant considered in this model is the so-called *Vehicle Longitudinal Dynamics*.

Considering:



- $\dot{v}(t)$ the vehicle acceleration, expressed in $[m/s^2]$
- $v(t)$ the vehicle longitudinal speed, expressed in $[m/s]$
- m the vehicle mass, expressed in $[kg]$
- $F_x(t)$ the longitudinal force applied to the vehicle center of gravity, expressed in $[N]$
- $F_s(t)$ the longitudinal force applied to the wheel on the terrain, expressed in $[N]$
- $F_d(v)$ the longitudinal force applied to the vehicle center of gravity due to the frictions with air and terrain, expressed in $[N]$
- I the moment of inertia of each one of the wheels, expressed in $[kg \cdot m^2]$
- r the radius of the wheel, expressed in $[m]$
- $\omega(t)$ is the angular speed of the wheel, expressed in $[rad/s]$
- $\omega_e(t)$ is the angular speed of the engine/electrical motor, expressed in $[rad/s]$
- $\omega_{eRPM}(t) = \frac{\omega_e(t) \cdot 60}{2\pi}$ is the angular speed of the engine/ electrical motor, expressed in [revolutions per minutes]
- $\dot{\omega}(t)$ is the angular acceleration of the wheel, expressed in $[rad/s^2]$
- S is the frontal surface of the car, expressed in $[m^2]$
- $c_x = 0.3$ is the automobile drag coefficient
- ρ is the average density of air at sea level in standard conditions $\rightarrow \rho = 1.25 \text{ kg/m}^3$
- i_g is the gearbox reduction ratio
- i_f is the final drive reduction ratio
- $i_t = i_g \cdot i_f$ is the total power train reduction ratio.

An extremely simplified model can be obtained as follow:

$$\dot{v}(t) = \frac{F_x(t) - F_d(v)}{m} \quad (1)$$

where $\dot{v}(t)$ is the vehicle acceleration, m is its mass, $F_x(t)$ is the longitudinal force applied to its center of gravity by the effects of the torque applied on the wheels, and $F_d(v)$ is the sum of the friction forces on the vehicle due to wheel-terrain and vehicle-air interactions.

Considering that the torque is equally split between the two wheels (valid only on straight tracks)

$$T(t) - 2 \cdot F_s(t) \cdot r = 2I \cdot \dot{\omega}(t) \quad (2)$$

the absence of slipping:

$$\begin{cases} 2 \cdot F_s(t) \cdot r = F_x(t) \\ v(t) = 2 \pi \cdot r \cdot \omega(t) \end{cases}$$

and considering the moment of inertia of the wheels $I = 0$, we can define the following equation, given that $T(t) = 2 \cdot F_s(t) \cdot r \rightarrow F_s(t) = \frac{T(t)}{2 \cdot r}$.

The drag force that limits the maximum speed of the vehicle is equal to:

$$F_d(v(t)) = X_{air} \cdot (v(t))^2 + X_{tyres} \cdot v(t) \quad (3)$$

where:

$$X_{air} = \frac{1}{2} \cdot S \cdot c_x \cdot \rho \quad (4)$$

and, as usually modeled:

$$X_{tyres}|_{X_{tyres}(50 \frac{km}{h})} = X_{air}(50 \frac{km}{h}) \rightarrow X_{tyres} = \frac{X_{air} \cdot 50}{3.6} \quad (5)$$

By substituting the (2) equation in (1), and by integrating both sides, we obtain:

$$\begin{aligned} v(t) &= \frac{1}{m} \int_0^t F_x(t) - F_d(v(t)) dt = \frac{1}{m} \int_0^t 2 \cdot F_s(t) - F_d(v(t)) dt = \\ &= \frac{1}{m} \int_0^t \frac{T(t)}{r} - F_d(v(t)) dt \quad (6) \end{aligned}$$

and, by substituting (3) in (6):

$$v(t) = \frac{1}{m} \int_0^t \frac{T(t)}{r} - X_{air} \cdot (v(t))^2 - X_{tyres} \cdot v(t) dt \quad (7)$$

Remember that the integrator block of Simulink requires an initial condition corresponding to the vehicle's longitudinal speed at the beginning of the simulation, $v(0)$. A possible configuration of the integration block is shown in Figure 2.

During the model development, put all the needed gain to obtain as an output of the physical model a speed expressed in km/h.

To simulate the slope θ of the terrain, it is possible to add the gravity force $F_g(\theta)$ as follows:

$$v(t) = \frac{1}{m} \int_0^t \frac{T(t)}{r} - X_{air} \cdot (v(t))^2 - X_{tyres} \cdot |v(t)| \cdot dt + m \cdot g \cdot \sin(\theta) \quad (8)$$

With $g = 9.81$ the gravity acceleration on Earth.

Reasonable values for an electric compact car can be:

- $m = 1600 \text{ kg}$
- $r = 0.3 \text{ m}$
- The torque T (at the wheel) can vary in the range $[-60; 960] \text{ Nm}$
- $S = 3.5 \text{ m}^2$
- $c_x = 0.3$

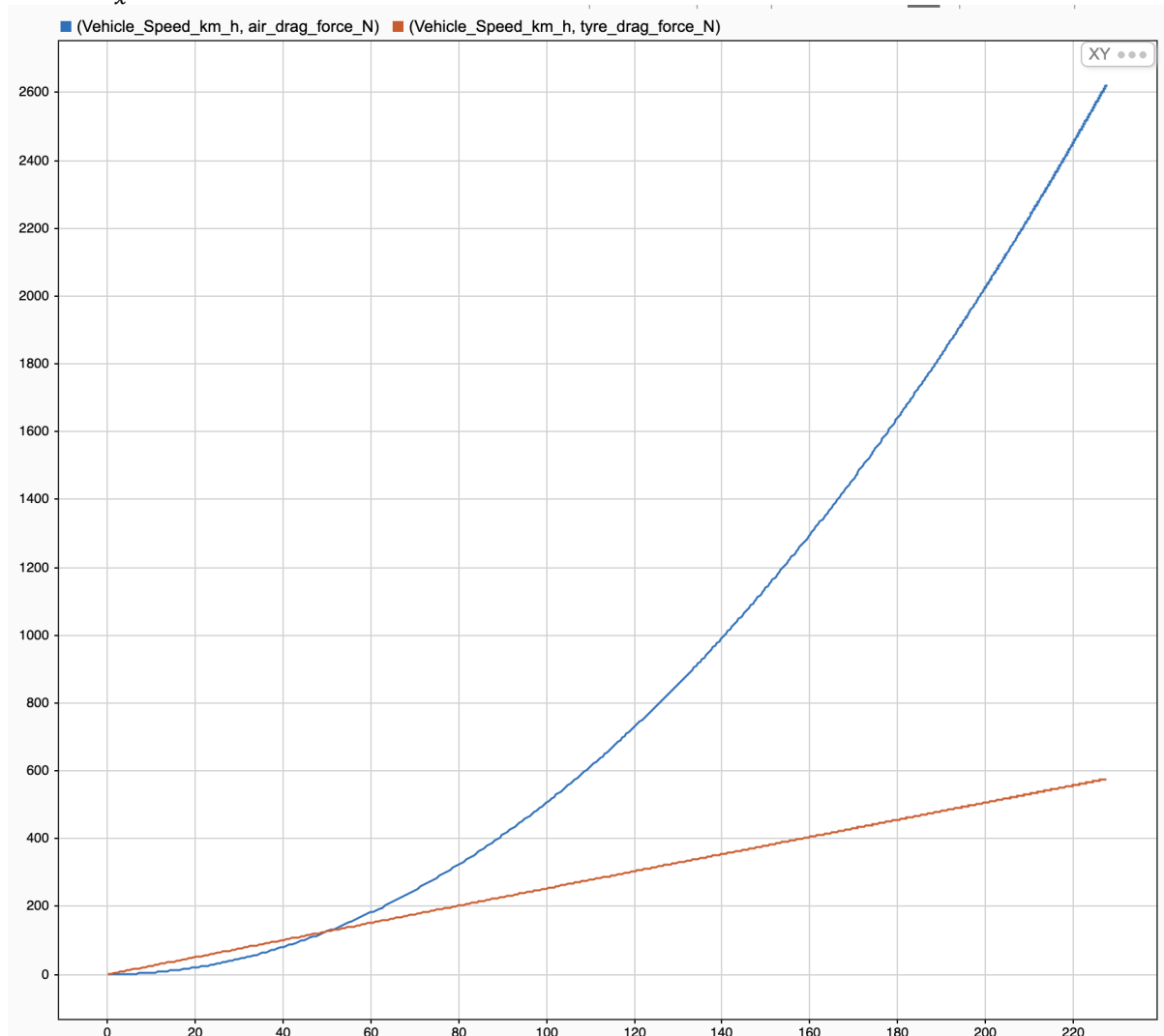


Figure 1 Graph showing drag forces of tires (in orange) and air (in blue) at various speeds. It is possible to observe that, as imposed in equation (5), $X_{\text{tyre}} = X_{\text{air}}$ at 50 km/h. Below this speed, the tire drag is dominant, after that, the air drag is dominant. Moreover, it is possible to see the top speed of the car (around 230 km/h) when $F_x = \frac{T}{r} = X_{\text{tyre}} + X_{\text{air}}$, with $F_x = 3200 \text{ N}$.

With those values, the top speed on level ground reachable by the car is about 230 km/h, where the drag forces equal the traction force (3200 N).

Considering the reverse direction, the maximum speed reachable with a limitation of -60 Nm is about 45 km/h.

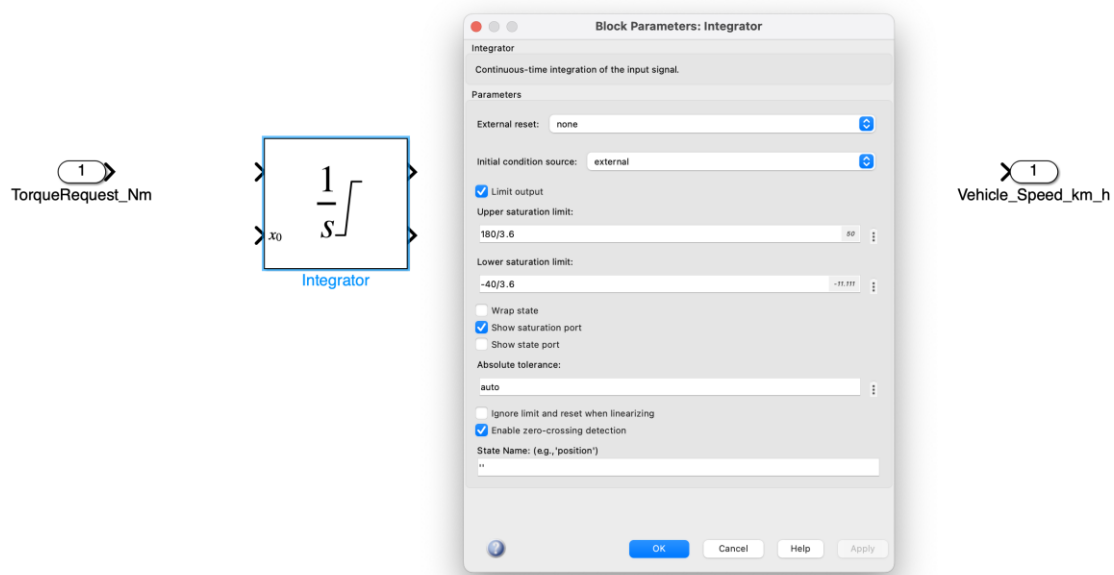


Figure 2 Settings window for the Integrator block of Simulink

Use these values (with a certain tolerance, for example, 10 %) to saturate the integrator block.

To make the model more realistic, it is possible to compute the torque request at the engine/motor. A typical ratio value for transmission of an electric car with a single gear can be around $i_t = 12^1$.

All the initialization parameters of the model are automatically loaded model by a callback of the function **init_fn** as shown in Figure 3.

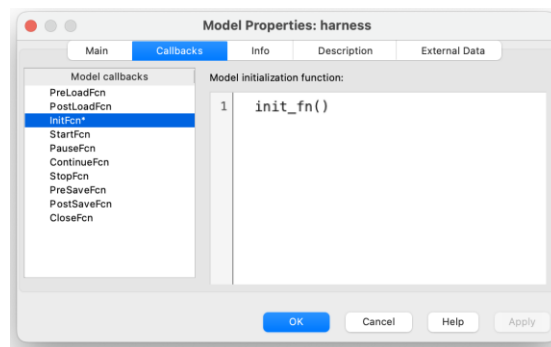
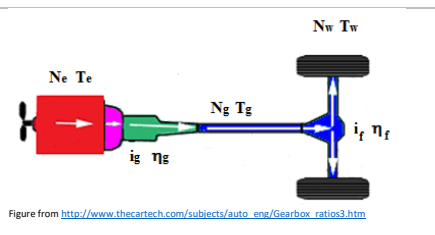


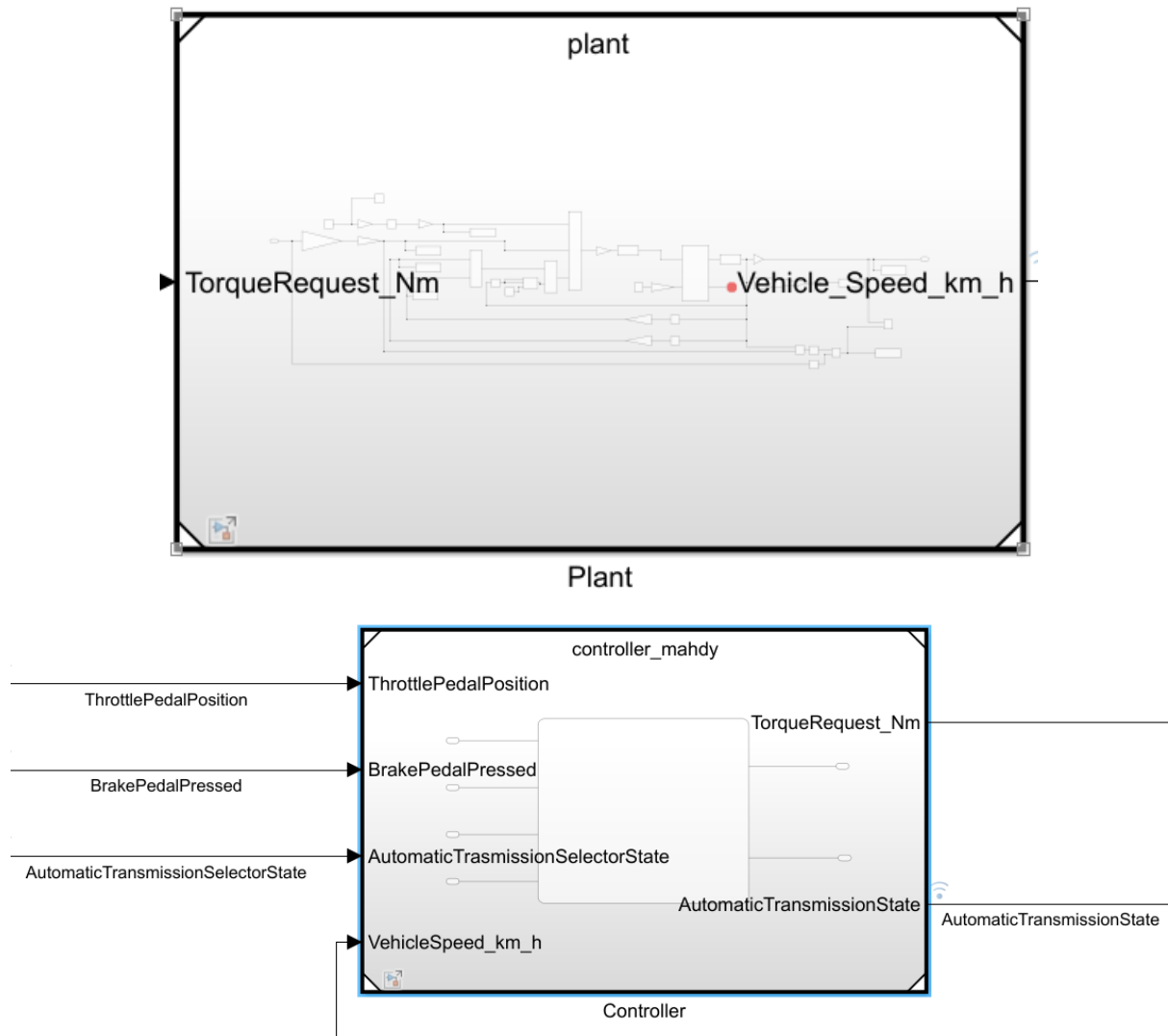
Figure 3 **init_fn** callback configuration in the harness model properties.

¹ Usually, the first gear of a car has a transmission ratio about $i_g = 3$, hence the engine makes 3 complete rotations very single rotation of the pinion gear of the differential. In the differential (final drive), the pinion gear makes about $i_f = 4$ revolutions for every revolution of the axle shafts. From here, the typical ratio of $i_t = 12$ between the torque at the wheel and the torque at the engine/electrical motor.



Description of the whole system

Draw the I/O block diagram of the plant and the controller, showing how they interact.



In this dynamical model, our harness consists of a driver where all the requested inputs are to the controller computes the torque required based on the throttle pedal position, brake pedal position as well as the automatic transmission state selector and velocity of the vehicle. This output is fed to the plant (the car itself in our case), that was built realizing the physical model and the vehicle longitudinal dynamics described above.

Controller SW Unit specifications

Provide a brief description of the Controller functionalities and its interfaces.

Our control will be responsible for selecting the appropriate state to which a torque is linked. Following the scheme the state of our system will be determined starting from a set of variables some of which selected by the driver and other calculated by our feedback loop specifically the velocity.

Interfaces

Name	Unit	Type ²	Data Type ³	Dimension	Min	Max
BrakePedalPressed	N/A	Input	Boolean	1x1	0	1
ThrottlePedalPosition	N/A	Input	Integer	1x1	0	1
AutomaticTransmissionSelectorState	N/A	Input	Enum	1x1	0	4
VehicleSpeed_kmh	Km/h	Input	Integer	1x1	-60	240
AutomaticTransmissionState	N/A	Output	Enum	1x1	0	4
TorqueRequest_Nm	Nm	Output	Integer	1x1	-80	80
Maximum_torque	Nm	Local	Integer	1x1	0	80
Maximum_torque in reverse	Nm	Local	Integer	1x1	0	40

Draw the Finite State Machine (FSM) representing the controller logic

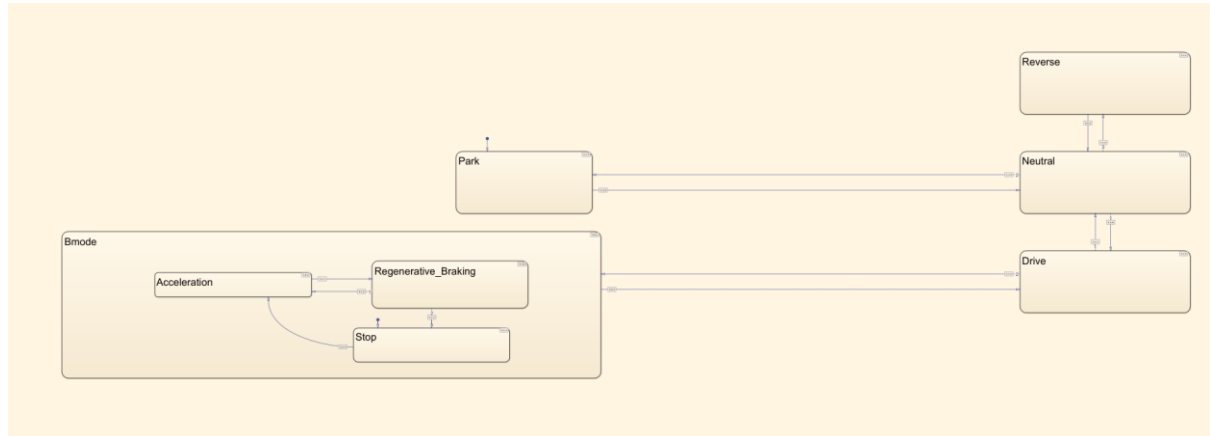


Figure 4: Controller of our system consisting of the state flow diagram

Comment on the design choices of the FSM, which are not trivial to be understood just by analyzing the controller logic.

This FSM design for the automatic transmission system includes several thoughtful, non-trivial choices that reflect deep considerations beyond basic controller logic. For instance, transitions out of the park state require the brake pedal to be pressed and vehicle speed to be

² Input, Output, Local, Global, Volatile

³ Struct, Double, Integer, Enum, Boolean, etc...

near zero, which is a deliberate safety feature to prevent accidental gear shifts while the vehicle is in motion. Additionally, the use of entry actions in each state to assign torque ensures that torque control is clearly encapsulated within each mode, avoiding cross-state interference and enhancing modularity. The duplication of similar transition conditions across states, rather than abstracting them into shared junctions, suggests a preference for explicit clarity over compactness, likely to aid traceability and debugging. Moreover, embedding complex logic like acceleration, regenerative braking, and stopping within the Brake mode as substates demonstrates a hierarchical structuring approach, which improves readability and reflects real-world vehicle behavior more accurately. These choices indicate a strong emphasis on safety, modularity, and transparency in the system's design.

Comment with plots of the results obtained from the test cases (it is suggested to use the Data Inspector)

The results shown in the plot from the Simulink Data Inspector confirm that the automatic transmission FSM behaves as expected under the test scenarios. As the gear selector (purple) changes from Park to Drive, Neutral, and back, the transmission state (blue) follows accordingly, indicating correct state transitions. When the throttle pedal (green) increases, the torque request (light blue) rises proportionally, validating the torque calculation logic in Drive mode. Later, when the throttle drops and the vehicle speed (aqua) decreases, the torque turns negative, demonstrating correct implementation of regenerative braking. The brake pedal input (orange) remains stable except during gear shifts, fulfilling safety conditions. Overall, the plots verify that the controller logic, torque behavior, and state transitions align with the FSM design and safety requirements.

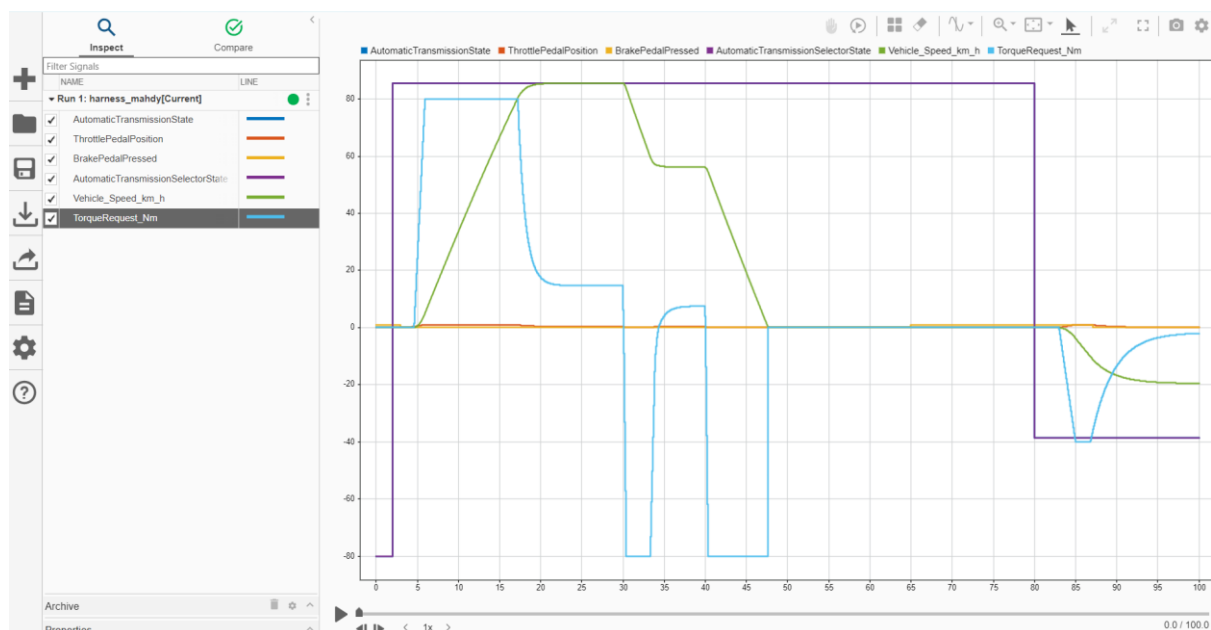


Figure 5: Final Results of our system stating all the system variables and relevant parameters